

# OS-Specific VPN Integration APIs & Requirements — Deep Research

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**Research Date:** July 2025

**Scope:** Native VPN APIs, network extension frameworks, and system-level integration requirements for designing Rust shared core platform adapters

**Protocols Covered:** WireGuard, OpenVPN, IKEv2/IPsec

**Platforms:** macOS/iOS, Android, Windows, Linux, HarmonyOS, Aurora OS (Sailfish OS)

**Searches Performed:** 12 independent web search queries across 60+ sources

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# 1. Executive Summary

This research provides a comprehensive analysis of the native VPN integration APIs across six operating system platforms, with focus on designing platform adapters for a Rust-based VPN shared core. Each OS presents a fundamentally different architecture for VPN integration, with varying capabilities, limitations, and certification requirements.

## Key Findings at a Glance

| Platform        | Primary API                               | Custom Protocol Support | Kill Switch Support                 | Split Tunneling                           |
|-----------------|-------------------------------------------|-------------------------|-------------------------------------|-------------------------------------------|
| macOS/iOS       | NetworkExtension (NEPacketTunnelProvider) | Yes (packet-level)      | PF firewall (macOS), Limited (iOS)  | Per-app via includedRouted excludedRoutes |
| Android         | VpnService + Builder                      | Yes (IP packet tunnel)  | Always-on + lockdown mode (API 24+) | Per-app (allowed/disallowed apps, routes) |
| Windows         | WFP callouts + WinTUN/NDIS                | Yes (via WinTUN driver) | WFP firewall rules                  | IP-based routing                          |
| Linux           | TUN/TAP + Netlink/NetworkManager          | Full (raw TUN device)   | nftables/iptables                   | Route-based cgroup-bpf                    |
| HarmonyOS       | VpnExtensionAbility + @ohos.net.vpn       | Limited (via TUN fd)    | Built-in isBlocking flag            | Per-app (trusted/blocked apps)            |
| Aurora/Sailfish | ConnMan VPN plugin + D-Bus                | Via wg-quick/userspace  | iptables/nftables                   | Route-based                               |

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## 2. macOS / iOS — NetworkExtension Framework

### 2.1 Architecture Overview

Apple's NetworkExtension framework is the sole sanctioned mechanism for building VPN applications on macOS and iOS. It provides deep system integration through a privileged app extension model.

The framework distinguishes between two main VPN approaches [^64^]:

**Personal VPN** (NEVPNManager): For standard VPN connections using IPsec or IKEv2 protocols. Leverages Apple's built-in VPN client infrastructure. Suitable for consumer-facing VPN services.

**Network Extensions** (NETunnelProviderManager): For implementing custom VPN protocols (WireGuard, OpenVPN, etc.) that are not IPsec/IKEv2. Uses NEPacketTunnelProvider for packet-level tunneling or NEAppProxyProvider for flow-based proxying.

### 2.2 Core Classes

| Class                    | Purpose                                                                 | Platform      |
|--------------------------|-------------------------------------------------------------------------|---------------|
| NEPacketTunnelProvider   | Main tunnel logic — receives raw IP packets, manages virtual interface  | macOS/<br>iOS |
| NETunnelProviderManager  | Manages VPN configuration in system preferences                         | macOS/<br>iOS |
| NETunnelProviderProtocol | Defines protocol-specific settings (server address, provider bundle ID) | macOS/<br>iOS |
| NEVPNManager             | Personal VPN management for IPsec/IKEv2 only                            | macOS/<br>iOS |
| NEOnDemandRule           | Rules for automatic VPN activation                                      | macOS/<br>iOS |

### 2.3 NEPacketTunnelProvider Lifecycle

The NEPacketTunnelProvider lifecycle follows a well-defined pattern [^142^]:

```

class PacketTunnelProvider: NEPacketTunnelProvider {
    override func startTunnel(options: [String : NSObject]?,
                               completionHandler: @escaping
                                   (Error?) -> Void)
    override func stopTunnel(with reason: NEProviderStopReason,
                               completionHandler: @escaping () ->
                                   Void)
    override func handleAppMessage(_ messageData: Data,
                                     completionHandler: ((Data?) ->
                                         Void)?)
    override func sleep(completionHandler: @escaping () -> Void)
    override func wake()
}

```

### Key lifecycle characteristics [^142^]:

- `startTunnel()`: Called when the main app invokes `startVPNTunnel()`. Must call `completionHandler(nil)` on success or `completionHandler(error)` on failure.
- `stopTunnel()`: Called on disconnect. Must call `completionHandler()` when cleanup is complete.
- `cancelTunnelWithError()`: Used internally by the provider to signal an error condition that should trigger reconnection.
- `sleep()/wake()`: Critical for handling device sleep/wake transitions. Must restore state properly.
- `handleAppMessage()`: IPC channel between the containing app and the extension via `NETunnelProviderSession.sendProviderMessage()`.

## 2.4 Entitlements & Provisioning

The Network Extension entitlement (`com.apple.developer.networking.networkextension`) is required [^70^]:

"Since November 2016, Packet Tunnel, App Proxy, Content Filter and DNS Proxy providers are self-serve — you enable the capability in Xcode or on the developer website with no request. Only Hotspot Helper and the NE app push provider remain managed."

```

<key>com.apple.developer.networking.networkextension</key>
<array>
    <string>packet-tunnel-provider</string>
</array>

```

**Critical deployment restriction [^64^]:** Provider types have different deployment restrictions. Some work only on **managed/supervised devices**. See Apple's [TN3134](#) for details.

## 2.5 Tunnel Configuration

Configuration is established through `NETunnelProviderManager` [^69^]:

```
let managers = try await
    NETunnelProviderManager.loadAllFromPreferences()
let manager = managers.first ?? NETunnelProviderManager()

let tunnelProtocol = NETunnelProviderProtocol()
tunnelProtocol.providerBundleIdentifier =
    config.providerBundleIdentifier
tunnelProtocol.serverAddress = config.serverAddress
tunnelProtocol.providerConfiguration = ["key": "value"]

manager.localizedDescription = config.displayName
manager.protocolConfiguration = tunnelProtocol
manager.isEnabled = true

try await manager.saveToPreferences()
```

**Important:** VPN configurations created using `NETunnelProviderManager` are classified as **enterprise VPN configurations**. Only one enterprise VPN can be enabled at a time. Enterprise VPN takes precedence over Personal VPN when routes conflict [^65^].

## 2.6 Packet Flow & Transport

`NEPacketTunnelProvider` provides two transport mechanisms for the tunnel [^129^]:

- **UDP:** `createUDPSession(to:)` — preferred for most VPN protocols (faster, lower overhead)
- **TCP:** `createTCPConnection(to:)` — useful for protocols requiring reliability or for traversing restrictive networks

"The primary API for implementing your custom tunneling protocols in `NEPacketTunnelProvider` allows you to tunnel traffic on an IP layer. They run as app extensions in the background handling network traffic." [^129^]

Packets flow through a virtual `utunX` interface. The system diverts IP packets to the `NEPacketTunnelProvider`, which encapsulates them and sends via UDP/TCP to the VPN server.

## 2.7 On-Demand Rules

Automatic VPN activation is supported via `NEOnDemandRule` subclasses [^64^]:

```
let onDemandRule = NEOnDemandRuleConnect()  
onDemandRule.interfaceTypeMatch = .wifi  
onDemandRule.ssidMatch = ["OfficeNetwork"]  
  
manager.onDemandRules = [onDemandRule]  
manager.isOnDemandEnabled = true
```

Also supports `NEOnDemandRuleEvaluateConnection` with `NEEvaluateConnectionRule` for domain-based triggering.

## 2.8 macOS-Specific: Kill Switch via PF

On macOS, kill switch is implemented using the **Packet Filter (PF)** firewall [^114^] [^219^]:

"IVPN has implemented a secure and robust mechanism called the IVPN firewall. Once enabled the IVPN Firewall integrates deep into the operating system (using Microsoft's own WFP API on Windows, pf on macOS, and iptables on Linux) and filters all network packets."

macOS 14+ PF anchors with block/allow policies:

```
block drop all  
pass on utunX all
```

**WWDC 2025 guidance** [^136^]: Apple explicitly discourages using Packet Filter or directly modifying routing tables on Mac. Network Extension is the supported API:

"Avoid using Packet Filter or directly modifying the routing table on the Mac. This is not supported and risk clashing with traffic filtering and routing rules installed by the system or other apps. If your VPN app doesn't use Network Extension today, you should migrate as soon as possible."

## 2.9 Key Limitations

| Limitation                                   | Details                                                                       |
|----------------------------------------------|-------------------------------------------------------------------------------|
| Only one enterprise VPN active               | System-wide limitation; conflicts with other VPN apps                         |
| iOS Simulator unsupported                    | Must test on physical devices                                                 |
| Extension memory limits                      | System may terminate under extreme memory pressure                            |
| No direct routing table modification (macOS) | Must use Network Extension APIs<br><br>Per-app proxy only for managed devices |

| Limitation                                    | Details                                                                            |
|-----------------------------------------------|------------------------------------------------------------------------------------|
| NEAppProxyProvider requires supervised device |                                                                                    |
| utun interface number changes                 | Must handle dynamic interface numbering after sleep/wake                           |
| iOS kill switch limited                       | No true kernel-level kill switch; relies on disconnectOnSleep and profile behavior |

## 3. Android — VpnService API

### 3.1 Architecture Overview

Android's VpnService is the core API for building custom VPN clients. It operates by creating a virtual network interface (TUN) and intercepting all IP traffic from the device.

### 3.2 Core API: VpnService.Builder

The VpnService.Builder class configures the VPN tunnel [^109^]:

```
public class VpnService.Builder {
    public Builder addAddress(String address, int prefixLength)
    public Builder addRoute(String address, int
        prefixLength) // Route through VPN
    public Builder excludeRoute(IpPrefix
        prefix) // Exclude from VPN (API 33+)
    public Builder addDnsServer(String address)
    public Builder addSearchDomain(String domain)
    public Builder addAllowedApplication(String
        packageName) // Per-app VPN (include)
    public Builder addDisallowedApplication(String
        packageName) // Per-app VPN (exclude)
    public Builder
        allowBypass() // Allow
        apps to bypass VPN
    public Builder setBlocking(boolean blocking)
    public Builder setMtu(int mtu)
    public Builder setSession(String session)
    public ParcelFileDescriptor
        establish() // Create TUN
        interface
}
```

### 3.3 Always-On VPN & Lockdown Mode

Android 7.0+ (API 24) introduced **Always-on VPN** [^29^]:

"Android can start a VPN service when the device boots and keep it running. This feature is called *always-on VPN*."

Key characteristics:

- System starts/stops the VPN service automatically
- Persists across reboots and app updates
- VPN app can opt-out via `SERVICE_META_DATA_SUPPORTS_ALWAYS_ON` metadata set to false

**Lockdown mode** (`BLOCK_CONNECTIONS_WITHOUT_VPN`) [^26^]:

"When `lockdownEnabled` is true, all network traffic is blocked if the VPN is not connected. No traffic can leak to the open internet."

| Feature                       | API Level | Behavior                                              |
|-------------------------------|-----------|-------------------------------------------------------|
| Always-on VPN                 | 24+       | Auto-starts VPN on boot, keeps it running             |
| Block connections without VPN | 24+       | System-level kill switch — blocks all non-VPN traffic |
| Lockdown exemptions           | 29+       | Specific apps can be exempted from lockdown           |
| User can't disable always-on  | 30+       | Admin-configured always-on cannot be disabled by user |
| System VPN app exclusion      | 34+       | System apps can be excluded from VPN                  |

### 3.4 Split Tunneling

Android supports split tunneling through multiple mechanisms [^109^] [^21^]:

**Route-based** (all Android versions):

```
builder.addRoute("0.0.0.0", 0) // Route all traffic through VPN
```

**Exclude routes** (API 33+):

```
builder.excludeRoute(IpPrefix("192.168.1.0", 24)) // Exclude LAN
```

**Per-app VPN:**



```
builder.addAllowedApplication("com.example.app")    // Only this
    app uses VPN
// OR
builder.addDisallowedApplication("com.example.app") // This app
    bypasses VPN
```

"addAllowedApplication() and addDisallowedApplication() are mutually exclusive — you allowlist specific apps or blocklist them, not both." [^26^]

### Pre-API 33 workaround for exclude routes [^21^]:

"For Android 33+, we used builder.excludeRoute to exclude the desired IPs from the VPN. For versions below Android 33, we relied on builder.addRoute and included the required IP addresses calculated using the WireGuard AllowedIPs Calculator."

## 3.5 Background Execution & Foreground Services

VPN services must run as **foreground services** to avoid being killed [^175^]:

"Due to Android battery optimizations introduced in Android 8.0 (API level 26), background services have now some important limitations. Essentially, they are killed once the app is in background for a while."

### Critical Android 15 change [^177^]:

"Starting in Android 15, all of an app's foreground services share a **6-hour time limit**."

The recommended approach for long-running VPN services:

- Use dataSync foreground service type (required for Android 14+)
- Use **User-Initiated Data Transfer (UIDT) Jobs** for Android 14+ to bypass restrictions
- For Android 13 and below: rely on Foreground Services + WorkManager

```
<service android:name=".VpnService"
    android:foregroundServiceType="dataSync"
    android:permission="android.permission.BIND_VPN_SERVICE">
    <intent-filter>
        <action android:name="android.net.VpnService"/>
    </intent-filter>
</service>
```

## 3.6 Battery Optimization

Enterprise environments can exempt VPN apps from battery optimization via [^63^]:

- Knox Service Plugin BatteryOptimizationAllowlist policy
- addPackageToBatteryOptimizationWhiteList() API (Knox SDK)
- EMM-managed configuration

## 3.7 Protecting Sockets

The `VpnService.protect()` method allows specific sockets to bypass the VPN tunnel:

```
val socket = DatagramSocket()  
VpnService.protect(socket) // This socket goes directly to  
physical network
```

This is essential for the VPN's own transport connection to avoid routing loops.

## 3.8 Key Limitations

| Limitation                                       | Details                                  |
|--------------------------------------------------|------------------------------------------|
| No raw IP socket access                          | Must use VpnService TUN interface        |
| Multiple VPNs can't run simultaneously           | Only one active VPN at a time            |
| Foreground service notification required         | User-visible notification mandatory      |
| 6-hour limit on foreground services (Android 15) | Requires UIDT jobs for long-running VPNs |
| excludeRoute() only API 33+                      | Requires workaround on older versions    |
| Battery optimization may kill service            | User or admin must whitelist app         |
| HttpProxy over split tunnel doesn't work         | Proxy can't reach excluded destinations  |

## 4. Windows — Filtering Platform & NDIS

### 4.1 Architecture Overview

Windows provides multiple APIs for VPN integration, ranging from user-mode UWP plug-ins to kernel-mode WFP callout drivers.

### 4.2 Primary APIs

| API Layer                               | Technology                       | Use Case                                          |
|-----------------------------------------|----------------------------------|---------------------------------------------------|
| <b>Windows Filtering Platform (WFP)</b> | Kernel-mode callout drivers      | Kill switch, traffic filtering, packet inspection |
| <b>NDIS Lightweight Filter (LWF)</b>    | Kernel-mode driver               | Low-level packet interception                     |
| <b>UWP VPN Platform</b>                 | Windows.Networking.Vpn namespace | Store-distributed VPN apps                        |
| <b>RAS API</b>                          | User-mode (Win32)                | VPN profile management, dial-up                   |
| <b>WinTUN / WinTap</b>                  | Virtual network adapter drivers  | TUN/TAP device for packet tunneling               |

### 4.3 Windows Filtering Platform (WFP)

WFP is the primary mechanism for implementing kill switches and traffic filtering on Windows [^219^]:

"On Windows, the core mechanism is the Windows Filtering Platform (WFP). The VPN client installs a callout driver or uses system layers to filter at transport and network stack levels, marking VPN interface traffic and blocking all else."

WFP filtering layers relevant to VPN:

- FWPM\_LAYER\_OUTBOUND\_IPPACKET\_V4 / V6 — Outbound IP packets
- FWPM\_LAYER\_INBOUND\_IPPACKET\_V4 / V6 — Inbound IP packets
- FWPM\_LAYER\_STREAM\_V4 / V6 — TCP/UDP streams
- FWPM\_LAYER\_IPFORWARD\_V4 / V6 — Forwarding decisions

**Kill switch implementation** via WFP [^221^]:

"On Windows — Use WFP to implement a block everything rule, then provide a higher priority rule to allow on the tunnel interface."

## 4.4 WinTUN Driver

The WireGuard project provides **WinTUN**, a virtual TUN adapter for Windows:

"If an interface has only one peer, and that peer contains an Allowed IP in /0, then WireGuard enables a so-called 'kill-switch', which adds firewall rules to do the following: Packets from the tunnel service itself are permitted; Loopback packets are permitted; DHCP for IPv4 and IPv6 and NDP for IPv6 are permitted; All other packets are blocked." [^156^]

**Key behavior** [^156^]:

- WinTUN automatically creates kill-switch firewall rules when AllowedIPs = 0.0.0.0/0, ::/0
- Uses 0.0.0.0/1 + 128.0.0.0/1 split to avoid triggering kill-switch semantics when not desired
- DNS on port 53 is restricted to configured DNS servers only

## 4.5 UWP VPN Plug-in Framework

Microsoft provides a UWP VPN platform for Store apps [^199^]:

"The Windows UWP VPN platform handles all aspects of creating a virtual network adapter and adding it to the system along with appropriate metrics and Name Resolution Policy Table (NRPT) entries, DNS servers, and more."

Key features:

- System tray integration
- Custom XML configuration (MDM/Intune push)
- Web authentication support (VpnForegroundActivatedEventArgs)
- Route and traffic filter configuration via StartWithTrafficFilter
- Packet encapsulation/decapsulation in background task

## 4.6 NDIS vs WFP

Community guidance from OSR Developer Community [^73^]:

"On NT6 use WFP unless your packet processing requires you to deal with MAC layer frames. Since you mention VPN and the only transport left is IP, I presume you mean an IPv4/6 VPN. So WFP will be just what you want."

**Modern recommendation:** For Windows 10/11, use WFP exclusively to maintain compatibility with HVCI (Hypervisor-Protected Code Integrity).

## 4.7 Kill Switch Implementation

Professional VPN clients implement kill switches on Windows using [ ^219^ ]:

```
# Block all outbound
```

```
New-NetFirewallRule -Direction Outbound -Action Block -Profile Any
```

```
# Allow VPN interface
```

```
New-NetFirewallRule -Direction Outbound -Action Allow  
-InterfaceAlias "WireGuard Tunnel"
```

```
# Set interface metrics
```

```
Set-NetIPInterface -InterfaceAlias "VPN" -InterfaceMetric 1
```

Advanced clients use WFP callouts for faster, more reliable filtering.

## 4.8 Key Limitations

| Limitation                                   | Details                                            |
|----------------------------------------------|----------------------------------------------------|
| WFP callout drivers require kernel-mode code | Must be signed, complex development                |
| BFE (Base Filtering Engine) service restart  | Firewall rules may be lost; need restoration logic |
| Multiple filter drivers can conflict         | Need careful weight/layer positioning              |
| NDIS LWF drivers also require signing        | WHQL certification recommended                     |
| Interface metric management                  | Windows auto-metric can override VPN priority      |

# 5. Linux — TUN/TAP & NetworkManager

## 5.1 Architecture Overview

Linux offers the most flexible and open VPN integration environment. The standard approach uses the kernel's TUN/TAP virtual network interface driver combined with userspace management tools.

## 5.2 TUN/TAP Device Creation

TUN devices are created via the `/dev/net/tun` character device:

```
int fd = open("/dev/net/tun", O_RDWR);
struct ifreq ifr;
memset(&ifr, 0, sizeof(ifr));
ifr.ifr_flags = IFF_TUN | IFF_NO_PI;
strcpy(ifr.ifr_name, "tun0");
ioctl(fd, TUNSETIFF, (void *)&ifr);
```

**Required capabilities:** `CAP_NET_ADMIN` for creating TUN devices.  
Can be granted via file capabilities:

```
sudo setcap cap_net_admin,cap_net_raw=eip ./vpn-client
```

## 5.3 NetworkManager Integration

NetworkManager 1.16+ provides native WireGuard support [^67^]:

"NetworkManager provides a de facto standard API for configuring networking on the host. This allows different tools to integrate and interoperate — from cli, tui, GUI, to cockpit."

D-Bus API for VPN management:

```
# Create WireGuard connection
nmcli connection add type wireguard ifname wg0 con-name my-wg0

# Configure and activate
nmcli connection modify my-wg0 ipv4.method manual ipv4.addresses
    192.168.7.5/24
nmcli --show-secrets --ask connection up my-wg0
```

NetworkManager D-Bus interfaces for VPN:

- `org.freedesktop.NetworkManager.Device.WireGuard` — WireGuard device operations
- VPN plugin service files in `/usr/lib/NetworkManager/VPN/`

## 5.4 WireGuard Kernel Module

WireGuard has been in the Linux kernel since 5.6 [^71^]:

```
/lib/modules/5.10.6-1/kernel/drivers/net/wireguard/wireguard.ko.gz
```

This provides native, high-performance WireGuard tunneling without userspace implementations.

## 5.5 systemd-networkd Integration

Alternative to NetworkManager, systemd-networkd can manage VPN interfaces via `.netdev` and `.network` files:

```
# /etc/systemd/network/wg0.netdev
[NetDev]
Name=wg0
Kind=wireguard

[WireGuard]
PrivateKey=<base64-private-key>
ListenPort=51820

[WireGuardPeer]
PublicKey=<peer-public-key>
AllowedIPs=0.0.0.0/0,::/0
Endpoint=<server>:51820
```

## 5.6 Kill Switch Implementation (nftables)

Modern Linux kill switches use **nftables** (kernel 5.10+ recommended) [^219^]:

```
table inet vpn {
    chain output {
        type filter hook output priority 0; policy drop;
        oifname "wg0" accept
        oifname "lo" accept
        ct state established,related accept
        ip daddr <vpn-server-ip> accept # Allow VPN endpoint
        drop
    }
}
```

Policy routing with fwmark:

```
# Mark VPN traffic
ip rule add fwmark 0x1 table 100
ip route add default dev wg0 table 100
```

## 5.7 Split Tunneling (cgroup-bpf)

Advanced split tunneling uses **cgroup-bpf** for process-level filtering [^219^]:

"For advanced setups, cgroup-bpf (BPF\_CGROUP\_INET\_EGRESS) filters at the process level — letting, for example, your admin's ssh bypass the kill switch for debugging while everything else remains blocked."

## 5.8 Flatpak/Snap Sandboxing Implications

Sandboxed Linux app distributions present challenges for VPN clients:

| Sandboxing System | Network Access                   | TUN Device       | Workaround                                           |
|-------------------|----------------------------------|------------------|------------------------------------------------------|
| Flatpak           | Portal-based (NetworkMonitor)    | No direct access | Use <code>--device=all</code> or run outside sandbox |
| Snap              | network, network-bind plugs only | No TUN access    | Classic confinement or snapd plugin                  |

VPN clients distributed via Flatpak/Snap typically cannot create TUN devices due to sandbox restrictions. Common approaches:

- Distribute the UI via Flatpak, run the VPN daemon as a systemd service outside the sandbox
- Use `pkexec` for privilege escalation
- Request classic confinement (Snap) or `--device=all` (Flatpak)

## 5.9 Key Limitations

| Limitation                         | Details                                                      |
|------------------------------------|--------------------------------------------------------------|
| Requires root or CAP_NET_ADMIN     | Must grant capabilities                                      |
| Multiple VPN management systems    | NetworkManager vs systemd-networkd vs manual scripts         |
| iptables vs nftables fragmentation | iptables is deprecated; nftables is modern but not universal |
| No unified sandbox support         | Flatpak/Snap break TUN access                                |
| DNS leak prevention is manual      | Must configure systemd-resolved or resolv.conf management    |

# 6. HarmonyOS — VpnExtensionAbility

## 6.1 Architecture Overview

HarmonyOS provides a VPN framework through `VpnExtensionAbility` and the `@ohos.net.vpn` module. The architecture follows a model similar to Android's `VpnService` but with HarmonyOS-specific APIs.



## 6.2 VpnExtensionAbility

The VpnExtensionAbility class provides lifecycle callbacks for third-party VPNs [^77^]:

```
import { VpnExtensionAbility } from '@kit.NetworkKit';
import { Want } from '@kit.AbilityKit';

class MyVpnExtAbility extends VpnExtensionAbility {
  onCreate(want: Want) {
    console.info('MyVpnExtAbility onCreate');
  }
  onDestroy() {
    console.info('MyVpnExtAbility onDestroy');
  }
}
```

**Context:** VpnExtensionContext provides the VPN-specific runtime context.

## 6.3 VpnConnection API

The @ohos.net.vpn module provides VpnConnection for VPN management [^195^]:

```
import vpn from '@ohos.net.vpn';

// Create VPN connection
const VpnConnection = vpn.createVpnConnection(context);

// Set up VPN (returns TUN file descriptor)
let config: vpn.VpnConfig = {
  addresses: [{
    address: { address: "10.0.0.5", family: 1 },
    prefixLength: 24
  }],
  mtu: 1400,
  dnsAddresses: ["114.114.114.114"],
  isBlocking: true, // Kill switch
  trustedApplications: [], // Per-app split tunneling (include)
  blockedApplications: [] // Per-app split tunneling (exclude)
};

VpnConnection.setUp(config).then((tunfd: number) => {
  console.info("setUp success, tunfd: " + tunfd);
});

// Protect socket (bypass VPN)
VpnConnection.protect(socketFd);
```

```
// Destroy VPN
VpnConnection.destroy();
```

## 6.4 Permission Model

VPN functionality requires the `ohos.permission.MANAGE_VPN` permission [^197^]:

| Attribute          | Value                                   |
|--------------------|-----------------------------------------|
| Permission         | <code>ohos.permission.MANAGE_VPN</code> |
| Permission level   | <code>system_basic</code>               |
| Authorization mode | <code>system_grant</code>               |
| Enable via ACL     | <code>true</code> (API 12+)             |
| Valid since        | API 10                                  |

"Allows a system application to enable or disable the VPN function." [^197^]

**API 10-11:** Only system apps could use VPN APIs.

**API 12+:** ACL (Access Control List) enables third-party apps to request this permission.

## 6.5 VpnConfig Options

| Field                            | Type                                  | Required | Description                                  |
|----------------------------------|---------------------------------------|----------|----------------------------------------------|
| <code>addresses</code>           | <code>Array&lt;LinkAddress&gt;</code> | Yes      | IP address of the vNIC                       |
| <code>routes</code>              | <code>Array&lt;RouteInfo&gt;</code>   | No       | Route information                            |
| <code>dnsAddresses</code>        | <code>Array&lt;string&gt;</code>      | No       | DNS server IPs                               |
| <code>mtu</code>                 | <code>number</code>                   | No       | Maximum transmission unit                    |
| <code>isIPv4Accepted</code>      | <code>boolean</code>                  | No       | IPv4 support (default: true)                 |
| <code>isIPv6Accepted</code>      | <code>boolean</code>                  | No       | IPv6 support (default: false)                |
| <code>isBlocking</code>          | <code>boolean</code>                  | No       | Blocking mode / kill switch (default: false) |
| <code>trustedApplications</code> | <code>Array&lt;string&gt;</code>      | No       |                                              |

| Field               | Type          | Required | Description                           |
|---------------------|---------------|----------|---------------------------------------|
|                     |               |          | Bundle names of trusted apps          |
| blockedApplications | Array<string> | No       | Bundle names of blocked apps          |
| isLegacy            | boolean       | No       | Built-in VPN support (default: false) |

## 6.6 Key Limitations

| Limitation                            | Details                              |
|---------------------------------------|--------------------------------------|
| MANAGE_VPN is system-level permission | Requires system app or ACL (API 12+) |
| No direct raw socket access           | Must use provided socket APIs        |
| Third-party VPN is relatively new     | Ecosystem still maturing             |
| ArkTS/ArkUI native C interop          | May need NAPI for Rust integration   |

## 7. Aurora OS / Sailfish OS — ConnMan Integration

### 7.1 Architecture Overview

Aurora OS (Russian Sailfish OS fork) uses the same networking stack as Sailfish OS, centered on **ConnMan** (Connection Manager) as the system network manager. VPN integration is achieved through ConnMan VPN plugins.

### 7.2 ConnMan VPN Plugin Architecture

Sailfish OS uses ConnMan's VPN plugin system for VPN integration [^152^]:

"SailfishOS doesn't provide WireGuard functionality out-of-the-box, so we first need to install a few third-party programs from OpenRepos."

Required packages for WireGuard on Sailfish OS:

- wireguard-go — Userspace WireGuard implementation
- wireguard-tools — Configuration utilities
- connman-plugin-vpn-wireguard — ConnMan VPN plugin
- jolla-settings-networking-plugin-vpn-wireguard — Settings UI integration

## 7.3 VPN Plugin Structure

ConnMan VPN plugins communicate via **D-Bus** [^153^]:

net.connman.vpn — D-Bus service for VPN management  
net.connman.vpn.Connection — Individual VPN connection interface  
net.connman.vpn.Manager — VPN manager interface

Key D-Bus properties:

- State — Connection state (association, configuration, ready, failure)
- SplitRouting — Split tunneling enabled/disabled

## 7.4 Configuration

Sailfish OS 5.0+ includes built-in WireGuard support [^154^]:

```
devel-su pkcon install jolla-settings-networking-plugin-vpn-wireguard
systemctl restart connman
```

Configuration is done through the Settings > System > VPN UI, or by importing WireGuard configuration files.

## 7.5 Key Characteristics

| Aspect               | Details                                             |
|----------------------|-----------------------------------------------------|
| Network manager      | ConnMan                                             |
| VPN plugin API       | ConnMan VPN D-Bus API                               |
| Settings integration | Jolla Settings VPN plugins                          |
| DNS handling         | connmand handles all DNS (127.0.0.1) [^154^]        |
| Userspace fallback   | wireguard-go when kernel module unavailable [^145^] |
| Platform SDK         | Qt/QML based                                        |

## 7.6 Key Limitations

| Limitation                      | Details                                   |
|---------------------------------|-------------------------------------------|
| No kernel WireGuard module      | Uses userspace wireguard-go               |
| Third-party package ecosystem   | Relies on OpenRepos community packages    |
| ConnMan crashes reported        | Stability issues with VPN plugins [^153^] |
| Limited native development docs | Community documentation only              |

## 8. Permission Models & MDM Integration

### 8.1 iOS/macOS Permissions & MDM

| Aspect             | Details                                                                              |
|--------------------|--------------------------------------------------------------------------------------|
| App Store Review   | VPN apps face strict review; must justify Network Extension usage                    |
| Entitlement        | <code>com.apple.developer.networking.networkextension</code> (self-serve since 2016) |
| Supervised devices | Some features (NEAppProxyProvider, content filtering) require supervised mode        |
| MDM configuration  | <code>com.apple.vpn.managed</code> payload for profile-based VPN                     |
| Per-app VPN        | Supported via MDM for managed apps                                                   |

#### Enterprise VPN deployment [^70^]:

- VPN configurations pushed via Apple Configurator, Profile Manager, or third-party MDM
- On-demand rules can be pre-configured
- Personal VPN vs Enterprise VPN precedence rules apply

### 8.2 Android Permissions & MDM

| Aspect        | Details                                                                                                        |
|---------------|----------------------------------------------------------------------------------------------------------------|
| Permission    | <code>android.permission.BIND_VPN_SERVICE</code>                                                               |
| System dialog | User must grant VPN permission via system dialog (one-time per app)<br>Fully managed, work profile, COPE modes |

| Aspect                   | Details                                                                            |
|--------------------------|------------------------------------------------------------------------------------|
| MDM (Android Enterprise) |                                                                                    |
| Always-on VPN            | Configurable via MDM; cannot be disabled by user (API 30+)                         |
| Lockdown mode            | Admin-configurable; blocks all non-VPN traffic                                     |
| Per-app VPN              | Via <code>addAllowedApplication()</code> / <code>addDisallowedApplication()</code> |
| Battery optimization     | Admin can whitelist via EMM policies                                               |

#### Key MDM policies [^26^]:

- `DISALLOW_CONFIG_VPN` — Prevent user from changing VPN
- `vpnConfigDisabled` — Disable VPN configuration UI
- `lockdownEnabled` — Enable kill switch
- Battery optimization allowlist — Exempt VPN from Doze

## 8.3 Windows MDM Integration

| Aspect                         | Details                                                       |
|--------------------------------|---------------------------------------------------------------|
| MDM Support                    | Microsoft Intune, VMware Workspace ONE, etc.                  |
| VPN Profile                    | VPN configuration service provider (CSP)                      |
| Native VPN                     | Built-in IKEv2 client configurable via MDM                    |
| Third-party VPN                | UWP VPN plugins can receive MDM config via custom XML [^199^] |
| WEF (Windows Event Forwarding) | Can monitor VPN events                                        |

## 8.4 Linux MDM/Enterprise

| Aspect                        | Details                                                |
|-------------------------------|--------------------------------------------------------|
| NetworkManager                | D-Bus API for configuration management                 |
| <code>nmcli/nmtui</code>      | CLI/TUI tools for VPN configuration                    |
| <code>systemd-networkd</code> | <code>.netdev/.network</code> file-based configuration |
| Puppet/Ansible/Chef           | Common for enterprise Linux VPN deployment             |
| No unified MDM                | Unlike mobile platforms, no single MDM standard        |

## 8.5 HarmonyOS MDM

HarmonyOS supports MDM through the **MDM Kit** [^174^]:

"Adds support for enabling or disabling the SMS, mobile data, airplane mode, notification, and NFC features. Adds multiple user behavior restriction policies for PCs and 2-in-1 devices."

VPN configuration is part of enterprise device management via Endpoint Central and similar platforms [^176^]:

"Centrally manage device configurations like Wi-Fi, VPN, email, certificates, and more."

# 9. Kill Switch Implementation

## 9.1 Cross-Platform Kill Switch Approaches

| Platform  | Mechanism                          | Implementation                                         | Reliability         |
|-----------|------------------------------------|--------------------------------------------------------|---------------------|
| macOS     | PF (Packet Filter) firewall        | Kernel-level anchor rules; block all except utun       | High                |
| iOS       | Network Extension profile          | Limited; disconnectOnSleep, profile-based block        | Medium              |
| Android   | System lockdown mode               | BLOCK_CONNECTIONS_WITHOUT_VPN + Always-on VPN          | High (system-level) |
| Windows   | WFP firewall rules                 | Block outbound except VPN interface; WinTUN auto-rules | High                |
| Linux     | nftables/iptables + policy routing | Drop default; allow only tunnel interface              | High                |
| HarmonyOS | isBlocking in VpnConfig            | Built-in blocking mode                                 | Medium              |

## 9.2 macOS Kill Switch (PF)

"On macOS, the kill switch hinges on Network Extension (NE) and the built-in Packet Filter (PF). An NE client controls the utun tunnel interface, while PF anchors enforce strict policies: block all outbound traffic except on utunX and allowed tunnel services." [^219^]

Implementation:

```
block-policy drop
skip on lo0
anchor "vpn-killswitch" {
    block drop all
    pass on utun* all
}
```

## 9.3 iOS Kill Switch Limitations

iOS does **not** provide a true kernel-level kill switch for third-party VPN apps [^117^]:

"On iOS, it seems like a kill switch isn't even possible on any version."

Workarounds:

- Use Apple's Always-on VPN (requires MDM/supervised device)
- `NEVPNProtocol.disconnectOnSleep` can disconnect on sleep
- Some apps implement "Network Protection" features that are limited

## 9.4 Android Kill Switch (Lockdown Mode)

Android's system-level kill switch is the most reliable on mobile [^107^]:

"Android 8.0 and later includes a native kill switch feature called 'Always-on VPN' combined with 'Block connections without VPN'. This system-level option works with any VPN app and provides reliable protection without depending on the VPN developer's implementation."

```
// Programmatic detection (not control – user/system must enable)
val vpnManager = context.getSystemService(Context.VPN_SERVICE) as
    VpnManager
// Check if always-on is configured for this app
```



## 9.5 Windows Kill Switch (WFP)

"The approach: set a Block All outbound rule, then carve out exceptions allowing traffic via the VPN interface. Advanced clients go further — installing WFP callouts that drop packets before TCP stack processing and mark VPN packets to bypass accidental allow rules." [^219^]

WireGuard's built-in kill switch [^156^]:

"If an interface has only one peer, and that peer contains an Allowed IP in /0, then WireGuard enables a so-called 'kill-switch', which adds firewall rules to drop all traffic that is not travelling over the VPN."

## 9.6 Linux Kill Switch (nftables)

"By 2026, iptables is still around but outdated. We create an inet table called vpn, with input, forward, and output chains. Output policy is drop. We allow: established/related connections; interface wg0 (or tun0); DNS over wg0; optionally localhost." [^219^]

---

# 10. Split Tunneling

## 10.1 Platform Support Summary

| Platform  | Mechanism                                    | Granularity            | Per-App Support                                             |
|-----------|----------------------------------------------|------------------------|-------------------------------------------------------------|
| macOS/iOS | includedRoutes/<br>excludedRoutes            | IP/subnet              | Limited (iOS not possible per-app)                          |
| Android   | addRoute()/<br>excludeRoute() +<br>app lists | IP/subnet +<br>app     | Yes<br>(addAllowedApplication/<br>addDisallowedApplication) |
| Windows   | Route table + WFP<br>filters                 | IP/subnet +<br>process | Yes (WFP process-<br>based)                                 |
| Linux     | Policy routing +<br>cgroup-bpf               | IP/subnet +<br>process | Yes (cgroup-bpf)                                            |
| HarmonyOS | trustedApplications/<br>blockedApplications  | App                    | Yes                                                         |

## 10.2 iOS Split Tunneling Limitations

Mullvad documents this clearly [^111^]:

"Q: Will you add split tunneling to the iOS / iPadOS app?"

"A: It is not possible to do this currently."

iOS Network Extension provides NEIPv4Route and NEIPv6Route for route-based splitting only. Per-app split tunneling is not available for third-party VPNs.

## 10.3 Android Split Tunneling

Android provides the most comprehensive split tunneling on mobile [^109^]:

**Route-based** (all versions):

```
builder.addRoute("0.0.0.0", 0)           // Default through VPN
builder.addRoute("10.0.0.0", 8)          // Corporate network
                                         through VPN
```

**Exclude routes** (API 33+):

```
builder.excludeRoute(IpPrefix("192.168.0.0", 16)) // Exclude LAN
```

**Per-app** (all versions):

```
builder.addAllowedApplication("com.company.app") // Only
                                         these apps use VPN
builder.addDisallowedApplication("com.local.app") // These
                                         apps bypass VPN
```

## 10.4 Windows Split Tunneling

Windows split tunneling is typically implemented via:

1. **Route table management:** Add/remove routes for specific subnets
2. **WFP callout drivers:** Process-based routing decisions [^224^]

"A callout driver implementing policy-based routing for Windows, based on process name. Redirects TCP connections of a given process into a given network, despite of a default route." [^224^]

## 10.5 Linux Split Tunneling

Linux offers the most flexible split tunneling:

- **Route-based:** ip route commands for subnet-based routing
- **cgroup-bpf:** Process-level filtering via eBPF

- **Network namespaces:** Complete network isolation per process group
- 

## 11. Background Execution Constraints

### 11.1 iOS/macOS Background Execution

Network Extension providers run as **separate system processes**, independent of the hosting app [^64^]:

"Apple has engineered this framework to allow VPN connections to operate at a system level, independent of the lifecycle of the application that configured them."

- The extension is managed by the OS, not the app
- Persists even if the containing app is terminated
- May be terminated under extreme memory pressure
- Must handle `sleep()/wake()` events properly

### 11.2 Android Background Execution

| Android Version | Constraint                             | Impact on VPN                            |
|-----------------|----------------------------------------|------------------------------------------|
| 8.0 (API 26)    | Background service limits              | Must use foreground service              |
| 9.0 (API 28)    | App Standby Buckets                    | May restrict background VPN service      |
| 10 (API 29)     | Background location restrictions       | Affects location-aware VPN features      |
| 12 (API 31)     | Foreground service launch restrictions | Must use notification trampoline         |
| 14 (API 34)     | Foreground service type required       | Must declare <code>dataSync</code> type  |
| 15 (API 35)     | 6-hour foreground service limit        | Requires UIDT jobs for long-running VPNs |

**Mitigation strategies** [^177^]:

- Foreground service with `dataSync` type
- User-Initiated Data Transfer (UIDT) Jobs (API 34+)
- Battery optimization exemption
- MDM-managed always-on VPN

## 11.3 Windows Background Execution

VPN on Windows typically runs as a **Windows Service**:

- Can start automatically on boot
- Not subject to user session limitations
- WFP callout drivers are kernel-mode (always active)
- UWP VPN plugins run as background tasks

## 11.4 Linux Background Execution

VPN on Linux runs as a **systemd service**:

- systemd-networkd or NetworkManager manages interfaces
- WireGuard interfaces can be configured to auto-start
- No arbitrary execution time limits
- Persists across user sessions

## 11.5 HarmonyOS Background Execution

VpnExtensionAbility runs as a system ExtensionAbility:

- Lifecycle managed by the Ability runtime
- onCreate()/onDestroy() callbacks for setup/teardown
- AppServiceExtensionAbility (API 12+) for extended background service capabilities [^174^]

---

# 12. Raw Socket Access

## 12.1 Platform Comparison

| Platform  | Raw Socket Access                       | VPN Relevance                  | Requirements                 |
|-----------|-----------------------------------------|--------------------------------|------------------------------|
| Linux     | Full support via socket(AF_PACKET, ...) | Custom protocol framing        | CAP_NET_RAW capability       |
| macOS     | Supported (root or entitlement)         | Low-level packet capture       | Root or specific entitlement |
| iOS       | Not available to third-party apps       | Must use NetworkExtension APIs | N/A                          |
| Android   | Not available                           | Must use VpnService TUN        | N/A                          |
| Windows   | Supported (admin + driver)              | NDIS/WFP development           | Administrator privileges     |
| HarmonyOS |                                         |                                |                              |

| Platform | Raw Socket Access         | VPN Relevance             | Requirements          |
|----------|---------------------------|---------------------------|-----------------------|
|          | Limited (via socket APIs) | Use provided network APIs | MANAGE_VPN permission |

## 12.2 Linux Raw Socket Details

Linux provides full raw socket access with `CAP_NET_RAW` [^143^]:

"The very handy `CAP_NET_RAW` capability can be used to open raw sockets. Capabilities are applied on a per-file basis with the `setcap` command."

```
sudo setcap cap_net_admin,cap_net_raw=eip ./vpn-client
```

**Required capabilities for VPN** [^147^]:

- `CAP_NET_ADMIN` — Network interface configuration, routing tables
- `CAP_NET_RAW` — Raw and packet sockets

## 12.3 Mobile Platform Limitations

Both iOS and Android **do not** provide raw socket access to third-party apps:

- iOS: Must use `NEPacketTunnelProvider` with TUN abstraction
- Android: Must use `VpnService.Builder.establish()` to get TUN fd
- Custom protocol implementations must work over the provided TUN interface

## 12.4 Implications for Rust Shared Core

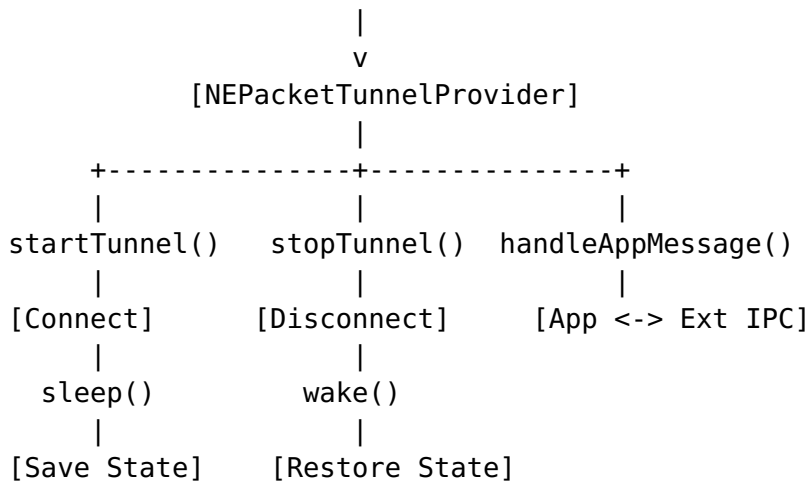
The Rust shared core should:

- Accept a TUN file descriptor (or platform equivalent) as the packet I/O channel
- Not attempt raw socket operations on mobile platforms
- Use platform-provided transport mechanisms (UDP/TCP sockets) for the VPN tunnel connection

# 13. Network Extension Lifecycle

## 13.1 iOS/macOS Lifecycle

[App Saves Config] -> [User Authorizes] -> [System Manages Extension]



### On-demand activation flow [^64^]:

1. Trigger condition met (Wi-Fi SSID match, domain lookup, etc.)
2. System automatically calls startTunnel()
3. Tunnel establishes
4. If condition no longer met, system may call stopTunnel()

## 13.2 Android Lifecycle

[App calls VpnService.prepare()] -> [System dialog] -> [Service.startForeground()]

[VpnService.run() - packet loop]

[Read packet] -> [Process]

[Read packet] -> [Write to TUN]

[Service.stopSelf() / System kills]

### Always-on lifecycle [^29^]:

1. Device boots
2. System starts VPN service automatically
3. System may restart service if it crashes
4. Service must detect system-initiated vs user-initiated starts

## 13.3 Windows Lifecycle

For service-based VPN:

[Service Auto-Start] -> [Create WinTUN Adapter] -> [WFP Rules Install]



### 13.4 Reconnection Handling

All platforms require careful reconnection handling:

| Platform  | Reconnect Mechanism                           | Key Consideration                            |
|-----------|-----------------------------------------------|----------------------------------------------|
| iOS       | cancelTunnelWithError() triggers system retry | Must preserve state in App Group             |
| Android   | System restarts service for always-on         | Service onStartCommand() must handle re-init |
| Windows   | Service control manager                       | Implement exponential backoff                |
| Linux     | systemd auto-restart                          | Restart=on-failure in unit file              |
| HarmonyOS | ExtensionAbility lifecycle                    | onCreate()/onDestroy() patterns              |

## 14. Cross-Platform Comparison Matrix

### 14.1 API × Capability Matrix

| Capability                     | macOS/iOS                                  | Android                  | Windows               | Linux                  |
|--------------------------------|--------------------------------------------|--------------------------|-----------------------|------------------------|
| Custom Protocol (packet-level) | Yes (NEPacketTunnelProvider)               | Yes (VpnService TUN)     | Yes (WinTUN)          | Yes (TUN fd)           |
| Custom Protocol (TCP/UDP)      | Yes (createUDPSession/createTCPConnection) | Yes (standard sockets)   | Yes (Winsock)         | Yes (Berkeley sockets) |
| System-managed persistence     | Yes (extension process)                    | Partial (foreground svc) | Yes (Windows Service) | Yes (systemd)          |
| Kill switch                    | PF (macOS) / Limited (iOS)                 | Always-on + lockdown     | WFP / WinTUN auto     | nftables               |
|                                | Yes                                        |                          | Yes                   | Yes                    |

| Capability                         | macOS/iOS                  | Android                    | Windows           | Linux       |
|------------------------------------|----------------------------|----------------------------|-------------------|-------------|
| <b>Split tunneling (route)</b>     | No (iOS) / Limited (macOS) | Yes (API 33+ excludeRoute) | Yes (WFP process) | Yes (cg)    |
| <b>Split tunneling (per-app)</b>   |                            | Yes (allowed/disallowed)   |                   |             |
| <b>On-demand rules</b>             | Yes (NEOnDemandRule)       | No                         | No                | No          |
| <b>MDM configuration</b>           | Yes (profile payload)      | Yes (AE policies)          | Yes (CSP)         | Partial     |
| <b>Raw socket access</b>           | Limited                    | No                         | Yes (admin)       | Yes (CAP_N) |
| <b>Battery optimization exempt</b> | N/A                        | Yes (API whitelist)        | N/A               | N/A         |

### 14.2 API × Limitation Matrix

| Limitation                         | macOS/iOS           | Android            | Windows           | Linux               | HarmonyOS        |
|------------------------------------|---------------------|--------------------|-------------------|---------------------|------------------|
| <b>Single VPN at a time</b>        | Yes (enterprise)    | Yes                | No                | No                  | Yes              |
| <b>Requires user authorization</b> | Yes (system dialog) | Yes (VPN dialog)   | No (service)      | No (root/caps)      | Yes (permission) |
| <b>Sandbox restrictions</b>        | App Extension model | Foreground service | UWP sandbox       | Flatpak/Snap        | Ability sandbox  |
| <b>Memory pressure kills</b>       | Yes (extension)     | Yes (background)   | No (service)      | No                  | Yes (Ability)    |
| <b>Custom protocol kernel code</b> | No                  | No                 | Yes (WFP callout) | Yes (kernel module) | No               |



# 15. Danger Zones & Platform-Specific Pitfalls

## 15.1 macOS/iOS Danger Zones

1. **PF routing table conflicts:** Apple explicitly warns against modifying routing tables directly [^136^]:

"Avoid using Packet Filter or directly modifying the routing table on the Mac. This is not supported and risk clashing with traffic filtering and routing rules installed by the system or other apps."
2. **utun interface number changes:** After sleep/wake, the interface number (e.g., utun3 -> utun4) can change. Must monitor and update firewall rules.
3. **Enterprise VPN precedence:** Only one enterprise VPN can be active. If multiple VPN apps are installed, they conflict.
4. **Extension memory limits:** System may terminate the packet tunnel provider under memory pressure. Must be efficient.
5. **iOS Simulator doesn't support Network Extensions:** Must test on physical hardware.
6. **App Store rejection risk:** VPN apps face heightened scrutiny. Must have clear privacy policy and legitimate use case.

## 15.2 Android Danger Zones

1. **Foreground service 6-hour limit (Android 15):** Critical for always-on VPN. Must migrate to UIDT jobs [^177^].
2. **Battery optimization:** OEM-specific "battery saver" features (Samsung, Xiaomi, etc.) can kill VPN services despite system settings.
3. **VPN permission revocation:** If user revokes VPN permission in Settings, the service is killed without warning.
4. **Multiple VPN apps:** Only one VPN can be active. Starting another VPN silently stops the current one.
5. **HttpProxy over split tunnel:** Using setHttpProxy() with split tunneling "generally won't work as expected" [^109^].
6. **excludeRoute() API 33+ only:** Must maintain dual code paths for route exclusion.

7. **Private Space (Android 15+):** VPN doesn't automatically apply to Private Space apps [^26^].

## 15.3 Windows Danger Zones

1. **BFE service restart:** If the Base Filtering Engine service restarts, WFP rules are lost. Must monitor and restore.
2. **HVCI compatibility:** Kernel-mode drivers must be compatible with Hypervisor-Protected Code Integrity on modern systems.
3. **Interface metric race:** Windows may override VPN interface metric, causing traffic to bypass the tunnel.
4. **Multiple filter drivers:** Conflicts between antivirus, VPN, and other WFP callout drivers are common.
5. **TDI filter deprecation:** Transport Driver Interface filters are deprecated; must use WFP.
6. **WireGuard kill-switch lockout:** Using AllowedIPs = 0.0.0.0/0 with the kill switch can block LAN access, including router admin interfaces [^151^].

## 15.4 Linux Danger Zones

1. **iptables vs nftables fragmentation:** Must support both or target specific distributions.
2. **DNS leak:** Linux doesn't automatically manage DNS when VPN connects. Must configure systemd-resolved or update resolv.conf.
3. **Flatpak/Snap sandbox:** TUN device access is blocked in sandboxed apps. Requires classic confinement or external daemon.
4. **Capability management:** CAP\_NET\_ADMIN must be set on the binary; easy to misconfigure.
5. **Route table conflicts:** Multiple VPN clients or network management tools can conflict on route table entries.
6. **WireGuard kernel module availability:** Not all distributions have the kernel module (especially older kernels).

## 15.5 HarmonyOS Danger Zones

1. **System API restriction:** MANAGE\_VPN was restricted to system apps until API 12. Third-party support is still maturing.

2. **ACL requirement:** Apps need Access Control List configuration for VPN permission.
3. **Native C interop:** ArkTS/ArkUI apps need NAPI for Rust integration, adding complexity.
4. **Ecosystem maturity:** VPN ecosystem on HarmonyOS is less mature than Android/iOS.

## 15.6 Aurora OS Danger Zones

1. **Userspace WireGuard only:** No kernel module; uses wireguard-go with performance implications.
  2. **Community package dependency:** Relies on OpenRepos for VPN packages, not official repositories.
  3. **ConnMan stability:** Reports of VPN plugin crashes and connection issues.
  4. **Limited documentation:** Mostly community-maintained documentation.
- 

## 16. Recommendations for Rust Shared Core Design

### 16.1 Platform Adapter Architecture

Based on this research, the Rust shared core should implement a **platform adapter pattern** with the following interfaces:

```
// Core abstraction for packet I/O
trait TunnelDevice {
    fn read_packet(&mut self, buf: &mut [u8]) -> Result<usize>;
    fn write_packet(&mut self, packet: &[u8]) -> Result<()>;
    fn set_mtu(&mut self, mtu: u16) -> Result<()>;
}
```

```
// Platform-specific VPN service lifecycle
trait PlatformAdapter {
    fn setup_tunnel(&self, config: TunnelConfig) -> Result<Box<dyn TunnelDevice>>;
    fn protect_socket(&self, socket: RawFd) -> Result<()>;
    fn configure_routes(&self, routes: &[Route]) -> Result<()>;
    fn configure_dns(&self, servers: &[IpAddr]) -> Result<()>;
    fn enable_kill_switch(&self) -> Result<()>;
    fn disable_kill_switch(&self) -> Result<()>;
}
```

```

    fn set_split_tunnel_apps(&self, allowed: &[String], blocked:
        &[String]) -> Result<()>;
}

```

## 16.2 Key Design Decisions

1. **Accept TUN fd from platform:** All platforms provide a TUN file descriptor. The Rust core should accept this fd and perform packet I/O through it.
2. **Separate transport from tunnel:** The platform adapter should handle the UDP/TCP transport socket to the VPN server (including `protect()`), while the Rust core handles packet encryption/decryption.
3. **Platform manages routing:** Route configuration should be handled by the native platform code (NetworkExtension on Apple, VpnService.Builder on Android, netlink on Linux, etc.).
4. **Kill switch delegated to platform:** Each platform has its own optimal kill switch mechanism. The Rust core should expose `enable/disable_kill_switch()` but the implementation should be platform-native.
5. **WireGuard in kernel space (Linux):** On Linux, use the kernel WireGuard module via Netlink when available, falling back to userspace implementation.
6. **Foreground service handling (Android):** The Android adapter must properly manage the foreground service lifecycle, notification, and Android 15 UIDT job migration.
7. **App Groups for state persistence (iOS):** Store connection state in App Group shared container for recovery after extension termination.

## 16.3 Implementation Priority

| Priority | Platform  | Rationale                                        |
|----------|-----------|--------------------------------------------------|
| P0       | Linux     | Most flexible, easiest to prototype              |
| P0       | Android   | Largest mobile market, well-documented API       |
| P0       | macOS     | NetworkExtension is mature and well-supported    |
| P1       | Windows   | Large desktop market, WFP complexity manageable  |
| P1       | iOS       | Similar to macOS but with additional constraints |
| P2       | HarmonyOS | Emerging market, API still maturing              |

| Priority | Platform  | Rationale                      |
|----------|-----------|--------------------------------|
| P2       | Aurora OS | Niche market, community-driven |

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*Research compiled from 12 independent web searches across 60+ authoritative sources including official developer documentation, open-source code repositories, security research papers, and industry technical blogs. All citations use [^number^] format with source URLs preserved.*